MIL-D-29362B(MC)
4 DECEMBER 1991
SUPERSEDING
MIL-D-29362A(MC)
13 April 1983

MILITARY SPECIFICATION

DECALS (HEAT TRANSFER): USMC

This specification is approved for use by the Marine Corps, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers requirements for face-down type decals required for marking service emblems on utility coats worn by Marine Corps personnel.

1.2 Classification. The decals shall be the Marine Corps emblem.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Commanding General (PSE-C), Marine Corps Research, Development, and Acquisition Command, Washington, DC 20380, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 7690

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SPECIFICATIONS

FEDERAL

P-S-1792 - Soap, Laundry (Neutral and Built)

STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures And Tables For Inspection by Attributes

MIL-STD-1188 - Commercial Packaging of Supplies and Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents and publications. The following other Government documents and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

LAWS AND REGULATIONS
U.S. Postal Service Manual

(Application for copies of the manual should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

2.2 Non-government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation.

UNIFORM CLASSIFICATION COMMITTEE, AGENT
Uniform Freight Classification Rules

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(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT
National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, Inc., Tariff Order Section, 1616 P. Street, N.W., Washington, D. C. 20026.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These douments also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Standard sample Standard samples are furnished solely for guidance and information to the contractor (see 6.3). Variation from this specification may appear in the standard sample in which case the specification shall govern.

3.2 First article. When specified (see 6.2), the contractor shall furnish one sample roll for the first article inspection and approval.

3.3 Materials.

3.3.1 Paper. Decals shall be printed on 4-1/2-inch-wide white transparent transfer paper, of standard commercial quality.

3.3.2 Printing medium. The decal markings shall be printed with a black, resin, printing medium which meets the colorfastness and printing requirements cited herein.

3.4 Design. The design of the decals shall be the United States Marine Corps design, to include size as shown on Figure 1.

3.5 Construction.

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3.5.1 Method of printing. Printing method shall be hot melt gravure process, offset (Lithographic) or letterpress, and shall be readily legible.

3.5.2 Put up. The decals shall be put up (wound) on tubes having an inside diameter of 2 inches and shall be furnished in rolls of 1000 decals per roll. There shall be an instruction/caution sheet conforming to Figure 2 inserted alternately between decals when put up in the roll form.

3.5.2.1 Perforations. Horizontal tearline perforations shall be printed and provided every 5 1/4 inches between decal and instructions, and shall be at a 90-degree angle to a straight line, bleeding off both sides, left and right. All perforations shall be cut clean and sufficiently deep to permit easy separation, but not to tear in ordinary handling.

3.5.3 Winding. The position of all decals, including instruction/caution sheet, on the roll, in relation to the direction in which the decals are wound on the tube, shall be in strict accordance with Figure 3. Roll paper must be tightly wound at even tension and shall not contain more than three splices per roll. Splices shall be neatly and securely overlapped/pasted with a repulpable adhesive which will not permit the splice to separate while passing through a drying oven maintained at 700°F. The adhesive may be applied from a tape form backing, providing the backing is removed, leaving only the adhesive component on the splice. The tails of the splices shall be neatly and evenly removed without damage to adjacent laps. The adhesive shall not cause the splice to adhere to adjacent laps. Splices shall be flagged at both ends of the roll with projecting colored markers, not pasted to the splice, or otherwise clearly marked. Roll width must be not less than specified herein and not more than 1/8 inch oversize.

3.6 Application. The decals shall be designed to be applied by a dry-heat transfer process so that initial print shall be well defined and clearly legible, shall not show smearing, bleeding, or offsetting and shall not show objectionable strike-through of the cloth to which applied.

3.7 Colorfastness. The decal markings, when applied by hand iron or automatic thermopress set at 300°F (for ironing or pressing cotton/linen items), shall show good colorfastness to laundering transferences, accelerated laundering, and color transference and shall be clearly legible when tested as specified in Section 4, Table I and 4.5.3.

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3.8 Workmanship. The finished decals shall be clean, the markings (printing) shall be well defined and clearly legible, shall show no smearing and shall conform to the quality and grade of production established by this specification. Ink coverage must be full and uniform, free from grayness, pinpoint and other imperfections. The occurrence of defects shall not exceed the applicable acceptable quality levels (AQL's).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certificate.

4 2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4 3).
b. Quality conformance inspection (see 4.4).
c. Packaging inspection (see 4.4.4).

4.3 Inspection. Inspection shall be performed in accordance with MIL-STD-105, except where otherwise indicated.

4.4 First article inspection. When first article is required, inspection and testing shall be made on completely fabricated decals for all provisions of this specification.

4.5 Quality conformance inspection.

4.5.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, standards, and drawings unless otherwise excluded, amended,modified or qualified in this specification or applicable purchase document.

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4.5.2 Examination of the end item. The end item shall be examined for the defects listed in 4.5.2.1. The sample unit for examination shall be one decal.

4.5.2.1 General defects. General defects shall be classified as follows:

Examine	Defect	Classification Major	Minor
Decals	a. Misshaped or distorted. b. Not specified size.	x	
Printing	a. Incorrect.	x	
	b. Omitted.	x	
	c. Illegible or smeared.	x	
	d. Wrong color (transfer paper or marking medium.)	x	
	e. Misplaced.		x
	f. Not specified size.	x	
	g. Spacing between decals not as specified.		x
	h. Print type not as specified.	x	

4.5.2.2 Inspection levels and AQLs. The required number of decals shall be inspected. The AQL shall be 2.5 defects per hundred units (DHU). The sample size shall be in accordance with level II of MIL-STD-105. The lot size shall be expressed in units of one decal of each type and class.

4.5.3 Testing of the end item. The method of testing specified in FED-STD-191, wherever applicable, and as listed in Table I shall be followed. The lot and sample unit size for each class of decal shall be as follows:

Lot size	Sample size
1000 up to and including 22,000	1 to 3 rolls
22,001 and over	5 rolls

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The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. In evaluating the printing component for colorfastness, the laundry transference (see Table I) shall be performed first. If the decal markings fail this test, no other tests shall be performed and the lot shall be rejected.

4.5.3.1 Test specimen. The specimen shall consist of a number of rectangles of cloth to which the heat transfer decals have been applied, weighing a total of 3.5 ± 0 3 grams (g). Each rectangle shall measure approximately 2 by 3 inches and shall be pinked along each edge. In addition, a thin ribbon of adhesive shall be applied to each pinked edge.

4.5.3.2 Apparatus.

4.5.3.2.1 Accelerator. The machine shall be capable of varying the speed of the rotor from 0 to 4,000 revolutions per minute (rpm) with a 4 1/2 inch "S" rotor. The liner for the test chamber shall be rubber ribbed, having two ribs per inch, with the ribs approximately 1/16 inch in height.

4.5.3.2.2 Soaking chamber. Any convenient vessel with 100 ml per g of specimen capacity.

4.5.3.2.3 Steam pressing. Flat bed press at temperatures of 135°C to 149°C (275° to 300°F), hot head or polished metal top for flat fabrics and cloth top press for rough crepes.

4.5.3.2.4 Hand pressing. A hand iron weighing approximately 6 pounds capable of maintaining temperatures between 135° to 149°C (275° to 300°F).

4.5.3.2.5 Soap. The soap shall conform to type IV of P-S-1792.

4.5.3.2.6 Adhesive. A latex acrylic adhesive shall be used for sealing edges of the cloth to which the decal markings have been applied to prevent fraying of the edges during the test cycle.

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TABLE I - Test Methods

Components	Characteristic	Rqmt para.	Test method	No. Determinations per Sample Unit	Results reported as
Transfer paper	Material identification	3.3.1	Std.Com'l 1/	-	-
Printing medium	Material identification	3.3.2	Std.Com'l 1/	-	-
Printing	Colorfastness to laundering transference	3.7	5610 2/	1	Pass or fail
	Colorfastness to transference	3.7	5640 3/	1	Pass or fail
	Colorfastness to accelerated laundering	3.7	4.5.3.3	1	Pass or fail

1/ Accepted on the basis of contractor's certificate of compliance.

2/ Method 5610 shall be followed except that in evaluating the results of this test, only the staining of color transfer cloth shall be evaluated.

3/ Method 5640 shall be followed except that the evaluation shall be conducted as follows:

a. When a standard sample has been established, the color transfer shall be evaluated as satisfactory or unsatisfactory by comparing the transfer produced by the standard sample with that produced by the specimen under examination.

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b. When no standard sample has been established, the transfer onto the transfer press cloth shall be rated as follows:

Excellent	:	No evidence of marking transferred to transfer cloth.
Good	:	No more than very slight transfer of marking to transfer cloth (i.e., required close examination).
Fair	:	Appreciable but not objectionable transfer of marking to the transfer cloth (i.e., marking is readily discernible upon examination).
Poor	:	Objectionable transfer of marking to transfer cloth.

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4.5.3.3 Procedure. The test procedures shall be as follows:

(a) The specimens shall be crumpled by hand to prevent their adhering to one another and then soaked for a period of approximately 30 ± 2 minutes at 150° ± 5°F in a soap solution of 100 milliliter (ml) per g of material.

(b) The specimens shall then be placed in the accelerator with 150 ml of the soap solution at 80° ± 2° F. The speed control of the accelerator shall be set to approximately 50 on the control dial. The motor shall then be started and quickly adjusted to 2000 rpm. (average speed shall be used since the tachometer pointer will fluctuate).

(c) The machine shall be run for exactly 6 minutes (including the time to adjust initial speed) without further adjustment of speed.

(d) The machine shall be shut down at the end of the 6-minute period and the liquid allowed to drain out. The specimens shall be removed, squeezed by hand and then rinsed free from soap with soft water at approximately 105° F. The exposed test specimens shall then be hydro-extracted or passed between wringer rolls to remove excess moisture and then dried by pressing between press cloths with a hand iron or flat bed press.

4.5.3.4 Evaluation. Change in color or definition of print shall be considered in rating colorfastness to laundering. The comparison shall be made with the test specimen and the standard sample, or comparison shall be made at normal reading distance under average north-sky daylight or equivalent light in the Procurement requirements.

Pass - Equal or superior to the standard sample.

Fail Inferior to the standard sample.

When no standard sample for comparison has been established, unless otherwise specified, the tested specimen shall be rated as to definition and legibility of print at a normal reading distance. Specimens shall be rated as follows:

Excellent - Practically no change in legibility or definition of print.
Good -Slight change, but readily legible.
Fair - Not readily legible, but without need for deciphering.
Poor -Not readily legible, requiring deciphering.

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4.5.4 Examination of packaging requirements. An examination shall be made to determine compliance with packaging requirements of Section 5. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packaged, selected just prior to closing operation. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the inspection lot. The inspection lot shall be S-2 and the AQL shall be 2 5 DHU.

Examine	Defect
Marking (exterior size, and interior)	Omitted; incorrect; illegible; of improper location, sequence or method of application.
Materials	Any component missing. Any component damaged, affecting serviceability
Workmanship	Inadequate application of component, such as incomplete closure of case liners, container flaps, loose strapping, inadequate stapling Bulging or distortion of containers
Content	Number of decals per shipping container is more or less than required

5. PACKAGING

5.1 Packaging. Packaging shall be commercial.

5.1.1 Commercial. Decals shall be packaged to afford adequate protection against deterioration and damage during shipment. The contractor may use his standard practice when it meets these requirements.

5.2 Packing. Packing shall be commercial.

5.2.1 Commercial. Decals, packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall comply with the U. S. Postal Service Manual, Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or purchase order, the shipping container shall be marked in accordance with the requirements of MIL STD 1188. A white label

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with the following information legibly printed or stamped thereon in black shall be inserted inside each tube in order to permit ready identification.

Nomenclature
Stock Number
Contract Number
Contractor

6. NOTES

6.1 Intended use. The decals are intended for use by Marine Corps personnel for marking utility uniforms.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(c) Whether first article is required (see 3.2).

6.3 Standard sample. For information regarding the availability of standard sample of the decal, address inquiry to the procuring activity issuing the invitation for bids.

6.4 First article. When a first article is required for inspection and approval, examination, tests and approval shall be as specified by the contracting officer.

6.5 Figures. Figures 1 through 3 are furnished for information only. In the event of conflict between the figures and the text of this specification, the text of this specification shall govern.

6.6 Changes from previous issues. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing Activity:
Navy-MC
Project No 7690-NO90

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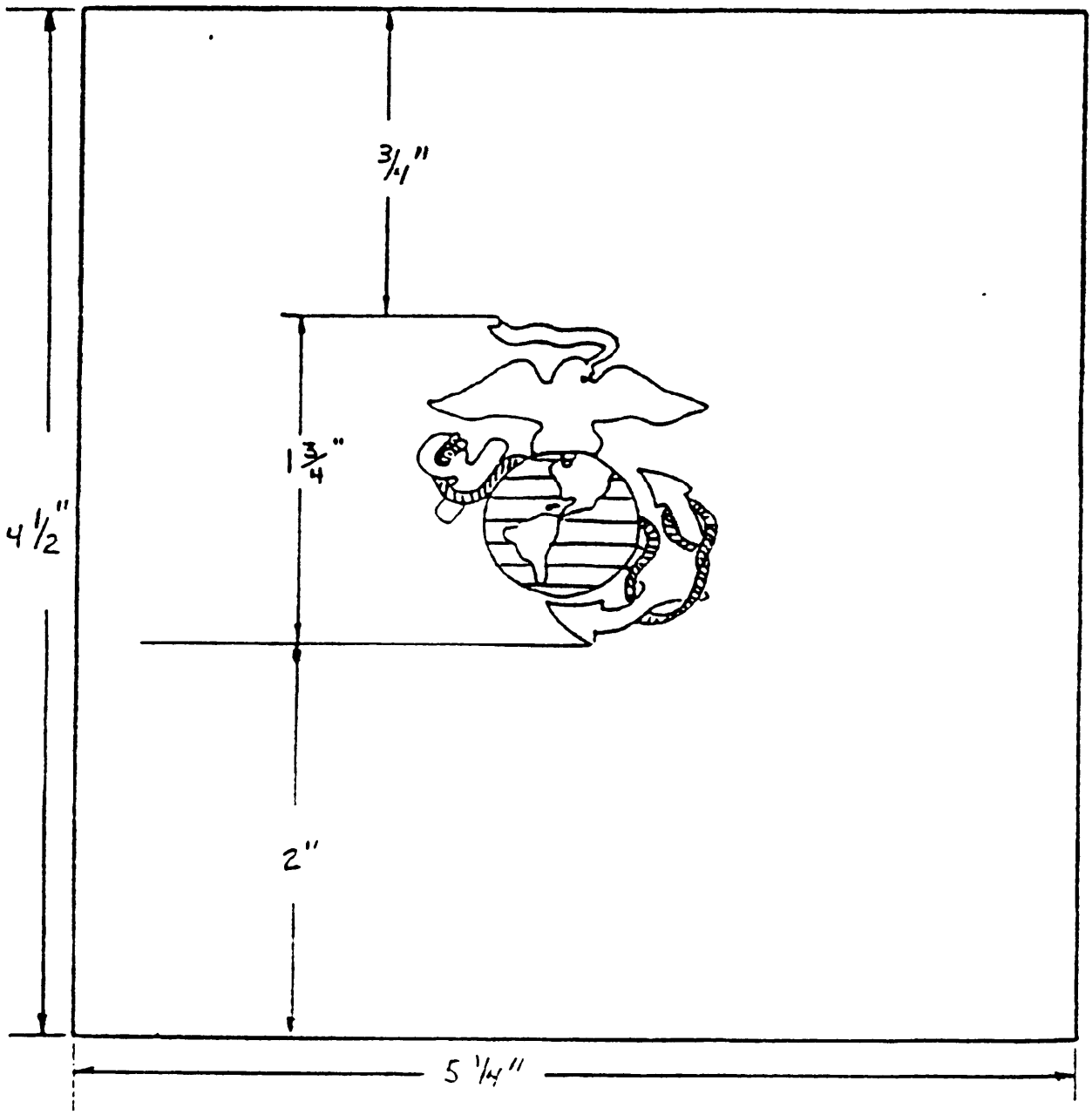
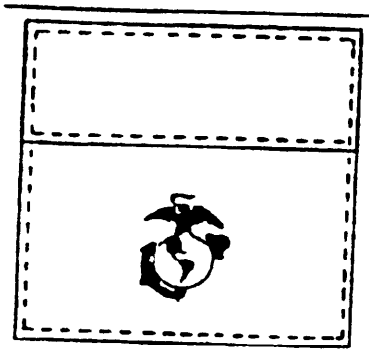


FIG:1 DECAL (HEAT TRANSFER)

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PLACEMENT ON POCKET



1. Detach instructions from decal.
2. Press pocket free of wrinkles.
3. Place decal on pocket with inked side down. Emblem/Letters centered on pocket.
4. Use dry iron set at 300° synthetic. Press over decal for 3 to 4 seconds. Do not use steam.
5. Remove paper immediately, while paper is hot. If paper should stick, reapply iron to decal and remove paper from pocket.

FIGURE 2 INSTRUCTION/CAUTION SHEET.

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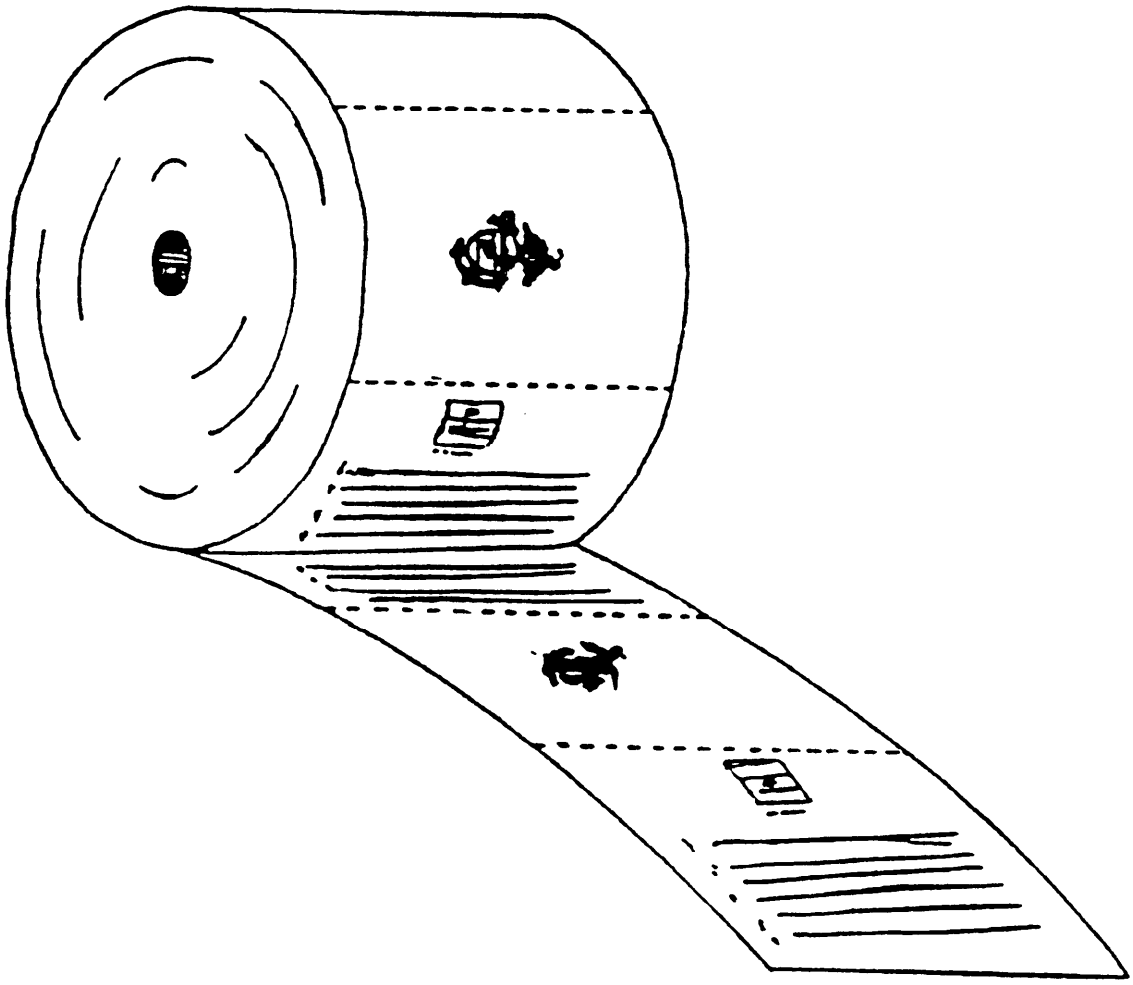


FIGURE 3 REQUIRED WINDING.

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
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3. DOCUMENT TITLE

DECALS (HEAT TRANSFER): USMC

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

[redacted]

6. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Commanding General, Marine Corps Research, Development, and Acquisition Command (PSE-C)	(202) 696-1186/87/88	226-1186/87/88
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:	
Washington, DC 20380-0001	Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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